Causal Attention and Causal Language Modeling

Part I: Autoregressive Prediction and the Causal Mask

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1 What is a causal language model?

A causal language model (causal LM or CLM) is trained to predict the next token using only the tokens that precede it. The word *causal* refers to the left-to-right factorization of the sequence probability.

For a sequence $x_1, \ldots, x_n$, the chain rule gives the autoregressive factorization

$$P(x_1, \ldots, x_n) = \prod_{t=1}^{n} P(x_t \mid x_1, \ldots, x_{t-1}).$$

For the first token, the context is empty or contains a beginning-of-sequence token. For every later position, the prediction depends on preceding tokens.

For example, given *The cat is*, the model predicts a possible next token such as *sleeping*. The prediction at the position of *is* must not use the future token *sleeping*.

2 How the causal attention mask works

Self-attention lets each token position gather information from other positions. Unmasked self-attention can use both earlier and later positions. For autoregressive prediction, future positions must be blocked.

Consider the sequence

$$\text{The} \quad | \quad \text{cat} \quad | \quad \text{is} \quad | \quad \text{sleeping.}$$

The following matrix indicates allowed attention. Rows are query positions (the token attending); columns are key positions (the token being attended to):

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

A 1 means attention is allowed; a 0 means it is blocked. Thus, *The* sees only itself; *cat* sees *The* and *cat*; *is* sees the first three tokens; and *sleeping* can see all four. In particular, the representation at *is* cannot use *sleeping*.

Scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + A\right)V,$$

where Q, K, V are query, key and value matrices. The additive mask is

$$A_{ij} = \begin{cases} 0, & j \le i, \\ -\infty, & j > i. \end{cases}$$

Conceptually, the negative infinity makes the softmax probability of a blocked future position zero. The mask is applied in the relevant self-attention layers, preventing future information from flowing into earlier token representations.

3 How next-token training works

A typical training sequence is shifted to form context-target pairs:

Context	**Target**
The	cat
The cat	is
The cat is	sleeping

A Transformer can process the whole input sequence in parallel. The causal mask ensures that each position uses only its allowed prefix, while the targets are shifted by one position. A common negative log-likelihood objective is

$$\mathcal{L}_{\text{CLM}} = -\sum_{t=1}^{n} \log P_\theta(x_t \mid x_1, \ldots, x_{t-1}).$$

Practical implementations handle beginning-of-sequence tokens, padding, and any positions excluded from the loss.

4 Does the causal mask alone make a model a causal LM?

No. Three things should be distinguished:

1. **Causal attention mask:** restricts which positions can exchange information.
2. **Next-token objective:** trains the model to predict the next token from the preceding context.
3. **Training arrangement:** aligns inputs and shifted targets correctly and uses the causal attention pattern.

A Transformer can use a causal mask but be trained to predict a class label or a numerical value. It then has causal attention, but its objective is not causal language modeling.

5 BERT and GPT

A GPT-style decoder-only model typically uses causal self-attention and next-token prediction. Together, these make it a causal LM.

Standard BERT, in contrast, uses bidirectional self-attention and is typically pretrained with masked language modeling. It can use both left and right context when constructing token representations, so standard BERT is not a causal LM.

Simply inserting a causal mask into pretrained BERT changes its information flow, but does not automatically turn its pretrained weights and objective into those of a properly trained causal LM. It would need an appropriate autoregressive training or adaptation procedure.

6 Main takeaway

The causal mask determines *what information a position may see*; the objective determines *what the model is trained to predict*. A typical causal language model combines both:

$$\boxed{\text{Causal attention} + \text{next-token prediction} \implies \text{causal language modeling.}}$$

The mask is an important mechanism for parallel autoregressive Transformer training, but it is not sufficient by itself.